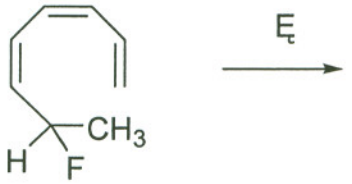
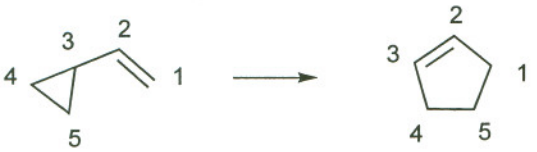
Organic Cume January 18, 2008 Pericyclic Reactions

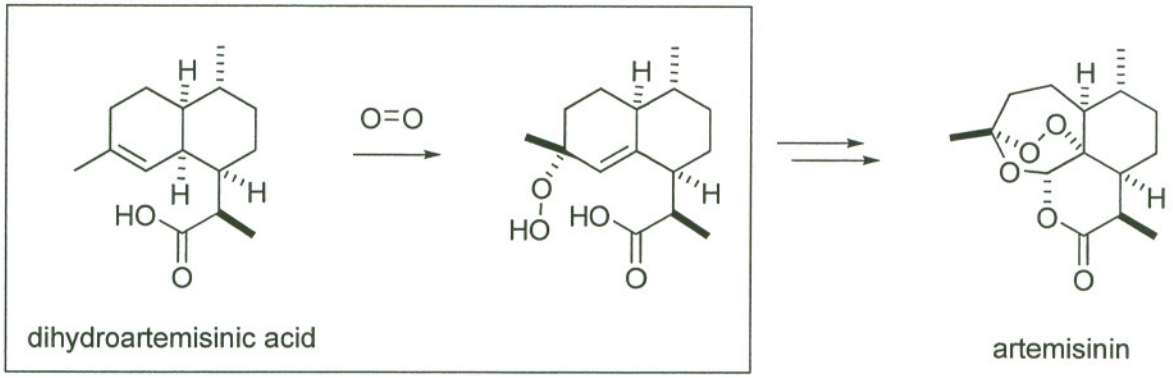
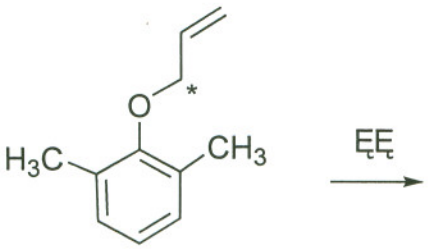
1. (a) Draw the product of an allowed thermal sigmatropic rearrangement of the hexatriene below (more than one answer is possible).
(b) Using the generalized orbital symmetry rule, designate the components of the allowed reaction. *Use the fewest number of components necessary to describe the process.*
(c) Draw the orbital interaction diagram for the reaction.



2. Consider the vinylcyclopropane to cyclopentene rearrangement shown below.
(a) Draw an orbital interaction diagram that would make this a thermally allowed reaction (more than one correct answer is possible).
(b) Experiment has shown that the reaction in fact occurs via a biradical mechanism. Based on your orbital interaction diagram, why do you think the reaction is biradical rather than pericyclic?



3. In a projected synthesis of artemisinin from dihydroartemisinic acid, the following thermal pericyclic reaction was used.
(a) Give the name of the reaction.
(b) Describe the reaction using orbital component nomenclature.



4. (a) Draw the intermediate(s) and aromatic product of the following reaction.
You don't need to draw electron pushing arrows or give any orbital descriptions.
(b) If the indicated (asterisked) carbon were labeled with ^{13}C, indicate with an asterisk where the label would be found in the final product.
(c) Give the name of any pericyclic reactions.